

alleles particular forms of genes

parents, such as color and height, could occur in all possible combinations in the offspring. Thus, a tall parent with green pods crossed with a short parent with yellow pods could give rise to offspring that were tall with yellow pods or short with green pods, as well as some of each parental type. Mendel concluded that the factors controlling height segregated independently from the factors controlling pod color. Later work showed that this was because these genes occurred on separate (nonhomologous) chromosomes, which themselves segregate independently during meiosis.

How is it possible for physically linked genes to nonetheless segregate independently? The answer lies in the events of crossing over. During crossing over, homologous chromosomes exchange segments at several sites along their length, in a process called recombination. Thus, two loci at distant ends of the chromosome are almost certain to have at least one exchange point occur between them. If only one exchange occurs, two alleles that began on the same chromosome will end up on different chromosomes. If there are two exchange points between them, they will end up together; if three, they end up apart, and so on. Over long distances, the likelihood of two **alleles** remaining together is only 50 percent, no better than chance, and, therefore, loci that are far apart on a large chromosome are not genetically linked. Conversely, loci that are close together will not segregate independently, and are therefore genetically linked. It is these that are most useful for mapping and discovering disease genes.

The loci examined in linkage analysis need not be genes of functional significance; indeed, anonymous segments of DNA (stretches of DNA with no known function) called genetic markers are often more useful in genetic linkage analysis. In order for a genetic marker to be of benefit in a linkage analysis, the chromosomal location of the marker must be known and, most importantly, there must be some variation in the sequence or length of these markers among individuals. Nongene markers used in linkage analysis are classified into four broad categories: restriction fragment length polymorphisms (RFLPs), variable number of tandem repeat (VNTRs), short tandem repeat polymorphisms (STRPs), and single nucleotide repeats (SNPs).

Calculating Linkage and Map Distance

As noted above, when genes are not genetically linked, alleles at the loci segregate independently from one another. So, if locus 1 has alleles A and a , and if locus 2, not linked to locus 1, has alleles B and b , then four gametes can be formed (AB , Ab , aB , and ab). Each of these four will occur with equal frequency (a 1:1:1:1 ratio), and all possible offspring combinations are expected with equal frequency.

If locus 1 and locus 2 are genetically linked to one another, however, deviations from this 1:1:1:1 ratio will be observed. If A and B begin on the same chromosome, then AB and ab will be more common than either aB or Ab . By counting the number of each type and determining the extent of this deviation, one can estimate the extent of recombination between the two loci: A large deviation means little recombination. The “recombination fraction,” expressed as a percentage, is an indirect measure of the distance between the loci and is the basis for the development of genetic maps.

Genetic maps order polymorphic markers by specifying the amount of recombination between markers, whereas physical maps quantify the distances among markers in terms of the number of base pairs of DNA. Although mapping in humans has a relatively recent history, the idea of a linear arrangement of genes on a chromosome was first proposed in 1911 by Thomas Hunt Morgan, who was studying the fruit fly, *Drosophila melanogaster*. The possibility of a genetic map was first formally investigated by the American geneticist Alfred H. Sturtevant in the 1930s, who determined the order of five markers on the X chromosome in *D. melanogaster* and then estimated the relative spacing among them.

For small recombination fractions (usually less than 10 percent to 12 percent), the estimate of the recombination fraction provides a very rough estimate of the physical distance. In general, 1 percent recombination is equivalent to about one million base pairs of DNA and is defined as one centimorgan. Physical measurements of DNA are often described in terms of thousands of **kilobases**. Crossing over does not occur equally at all locations, so estimates of distance from physical and genetic maps of the identical region may vary dramatically throughout the genome.

kilobases units of measure of the length of a nucleic acid chain; one kilobase is equal to 1,000 base pairs

Statistical Approaches

In experimental organisms, genetic mapping of loci involves counting the number of recombinant and nonrecombinant offspring of selected matings. Genetic mapping in humans is usually more complicated than in experimental organisms for many reasons, including researchers' inability to design specific matings of individuals, which limits the unequivocal assignment of recombinants and nonrecombinants. Therefore, maps of markers in humans are developed by means of one of several statistical algorithms used in computer programs.

Genetic maps can assume equal recombination between males and females, or they can allow for sex-specific differences in recombination, since it has been well established that there are substantial differences in recombination frequencies between men and women. Chromosomes recombine more often in females. On average, the female map is two times as long as the male map.

The complexity of the underlying statistical methods used to generate them renders genetic maps sensitive to marker genotyping errors, particularly in small intervals, and these maps are less useful in regions of less than about 2 centimorgans. While marker order is usually correct, genotyping errors can result in falsely inflated estimates of map distances.

Disease gene mapping is greatly facilitated by the availability of dense genetic maps. Linkage analysis for the mapping of disease genes boils down to the simple idea of counting recombinants and nonrecombinants, but in humans this process is complicated for a variety of reasons. The generation time is long in humans, so large, multigenerational **pedigrees** in which a disease or trait is segregating are rare. Scientists cannot dictate matings or exposures. They also cannot require that specific individuals participate in a study. Thus the process of linkage analysis in humans requires a statistical framework in which various **hypotheses** about the linkage of a trait locus and marker locus can be considered.

pedigrees sets of related individuals, or the graphic representation of their relationships

hypotheses testable statements



How far apart are the disease and marker, and how certain is the conclusion of linkage?

When the inheritance pattern for a disease is clearly known (e.g., autosomal dominant, sex-linked, etc.), the genetic data can be treated with a statistical approach that determines the likelihood that the gene is linked to a particular marker, at a particular position on a specific chromosome. This approach is often termed the “lod score approach,” where “lod” is short for logarithm of the odds.

Lod score linkage analysis is used most frequently to consider diseases that follow a Mendelian pattern of transmission within families. Positive lod scores, especially those greater than 3.0, suggest evidence for linkage between a disease gene and a marker locus. Negative lod scores suggest that the disease gene and marker locus are unlinked to one another. **SEE ALSO** CROSSING OVER; GENE DISCOVERY; HUMAN DISEASE GENES, IDENTIFICATION OF; MAPPING; MEIOSIS; MORGAN, THOMAS HUNT; POLYMORPHISMS.

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Maize

Maize (*Zea mays* L.), otherwise known as corn, is a highly unusual, economically important, and genetically well-characterized member of the grass family. It is believed to have originated some 8,000 to 10,000 years ago in the fields of the first agriculturalists of Mexico and Central America. These early farmers carefully selected traits that would ultimately transform the tiny, sparsely seeded spike of a wild grass into the large cob bearing many rows of kernels that we recognize today as an ear of corn.

The success of these early plant breeders was manifested by the spread of corn cultivation throughout the New World, long before the arrival of Europeans. Today, maize is grown in more countries than any other crop, and is a major source of food for both humans and domesticated animals throughout the world. The world production of maize in 2000 exceeded 23 billion bushels, the largest producer being the United States (43 percent).

Early Studies of Maize

As a major crop plant, maize was already the subject of study by plant breeders at the time of the rediscovery of Mendel's laws of inheritance at the beginning of the twentieth century. The inheritance patterns of readily observed traits were uncovered through controlled crosses and the examination of progeny. In many respects, maize was an ideal model system for this early period in the study of genetics. Male and female flowers are borne separately and are easily manipulated for controlled crosses. Large amounts of pollen are produced in the tassels (male inflorescence) over a period of days, and one ear (female inflorescence) contains many seeds (kernels). Large progeny arrays could be produced in one season.



Mutations caused differences in these ears of corn stored in the Maize Genetics Cooperation Stock Center, a repository for mutated maize located at the University of Illinois, Urbana-Champaign.

The high genetic diversity of maize provided many interesting mutant **phenotypes** to study, many of which were recessive. These could be maintained in a **heterozygous** state by the outcrossed breeding system (most fertilizations are the result of pollen transfer among plants) and easily uncovered by selfing (fertilizations that result from a plant's own pollen). There was also ample scope for selection of extreme phenotypes in continuous (quantitative) traits. A drawback for maize, compared to short-lived fruit flies, is that it only produces one or two crops per year, depending on location. However, many early maize geneticists knew that kernel phenotypes, which were discernable at harvest time, often predicted phenotypes in the adult plants, and could be used to set up the following season's crosses.

phenotypes observable characteristics of an organism

heterozygous characterized by possession of two different forms (alleles) of a particular gene



One of the earliest breakthroughs in crop breeding was the detection of hybrid vigor in maize by George Harrison Shull in 1908. He found that the progeny of two inbred lines were more productive than their wind-pollinated progenitors. This discovery provided the stimulus for the commercial propagation of maize and made it one of the most productive food plants worldwide.

Later Maize Studies

Many important genetic discoveries were made in maize by a group of scientists brought together at Cornell University in the 1920s and 30s by R. A. Emerson, who is often referred to as the spiritual father of maize genetics. The Emerson group, which included the future Nobel laureates Barbara McClintock and George Beadle, laid the foundation of maize genetics. They assembled information on maize mutants and ultimately produced the first genetic map of maize, based on linkage studies, in 1935. McClintock's first major contribution occurred early in her career (1929), when she perfected the techniques used to visualize maize chromosomes under the microscope. This allowed individual chromosomes to be identified by size, form, and features such as the highly staining regions, called "knobs."

This milestone allowed McClintock and other members of the Emerson group to make major advances in **cytogenetics**, which combines genetic crossing data and cytological landmarks to locate genes on chromosomes. Cytological landmarks include **trisomics**, reciprocal translocations, and deficiencies. Another of McClintock's breakthroughs, achieved with the collaboration of her colleague Harriet Creighton, was to establish the cytological proof of crossing over, which refers to the exchange of chromosomal segments during meiosis. Of course, McClintock's most famous discovery was that genetic elements within the genome can move (transpose) from one locus on the chromosome to another. These "jumping genes" (transposable genetic elements of transposons) were later discovered in bacteria, flies, and humans and eventually resulted in McClintock receiving a Nobel Prize in 1983.

In recent years, transposable elements have been exploited as tools for understanding the function of many maize genes. If a transposon inserts into a gene, it will disrupt the function of that gene. The disruption of gene function may result in a mutant phenotype affecting tissues or developmental stages of the plant that give some indication of the function of that gene. For instance, a transposon that inserts into a gene required for chlorophyll production would result in an albino seedling. Because the DNA sequences of many transposable elements in maize are known, they provide convenient molecular tags with which to clone and further characterize the gene into which they have inserted. Corn transposons have also been adapted to mutagenize and "tag" genes in the model plant *Arabidopsis thaliana*. SEE ALSO *ARABIDOPSIS THALIANA*; *CROSSING OVER*; *HETEROZYGOTE ADVANTAGE*; *MCCLINTOCK, BARBARA*; *MODEL ORGANISMS*; *TRANSPOSABLE GENETIC ELEMENTS*.

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cytogenetics study of chromosome structure and behavior

trisomics mutants with one extra chromosome

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Mapping

Genetic mapping is the process of measuring the distance between two or more **loci** on a chromosome. In order to determine this distance, a number of things must be done. First, the loci (pronounced "low-sigh") have to be known, and **alleles** have to exist at each locus so that they can be observed. The specific pair of alleles that are present is usually referred to as a genotype. Second, there has to be a way to measure the distance between the loci.

In genetic mapping, this distance is measured by the amount of meiotic recombination that occurs between the two loci. Meiotic recombination is the process in which the two chromosomes that are paired during meiosis each break apart and then reattach to each other, rather than back to themselves. These recombined chromosomes will end up in either eggs (for women) or sperm (for men).

Typically for any chromosome pair there will be only one or two such breaks per chromosome arm. The closer together two loci are, the less likely it is that such a break will occur. Thus, counting the number of breaks between two loci provides a good estimate of how far apart two loci are.

Genetic maps provide the order and distance between many markers all along the chromosome. In genetic maps, the loci that are used are called marker loci. Marker loci are almost always not in genes and serve only as signposts along the chromosome, "marking" a specific location. Thus genetic maps act much like road maps, and markers act much like mile markers or exit signs.

Why Create and Use Maps?

Genetic maps contain very important information and are used to help find the genes that can cause, or change the risk of developing, genetic diseases. For most diseases, the gene is not yet known and could be any one of the 30,000 to 70,000 genes that exist in the human genome. Since the disease gene is not known, its location is also not known. However, if the general location could be determined, then it would be much easier to figure out which of the genes near that location are the actual disease genes.

Genetic maps are very important for "disease-gene discovery," as they provide the reference locations for locating the disease gene. Finding the disease genes without a genetic map would be like trying to find a town by driving down a road without any mile markers or exit signs. There would be no clues as to where you are. The maps make it much easier to "navigate" the chromosomes.

loci sites on a chromosome (singular, locus)

alleles particular forms of genes



Using Recombination and Map Functions

Genetic maps are created by measuring the amount of recombination that occurs between two or more loci. The easiest way to do this is to use families with a large number of children, since this provides a large number of recombination events to look at. Scientists have collected a panel of forty such families, called the CEPH families (pronounced “sef,” from the French Centre d’Étude du Polymorphisme Humain—the Center for the Study of Human Polymorphisms). These families are measured (genotyped) for the variations at each locus, and the inheritance of each allele at each locus is compared.

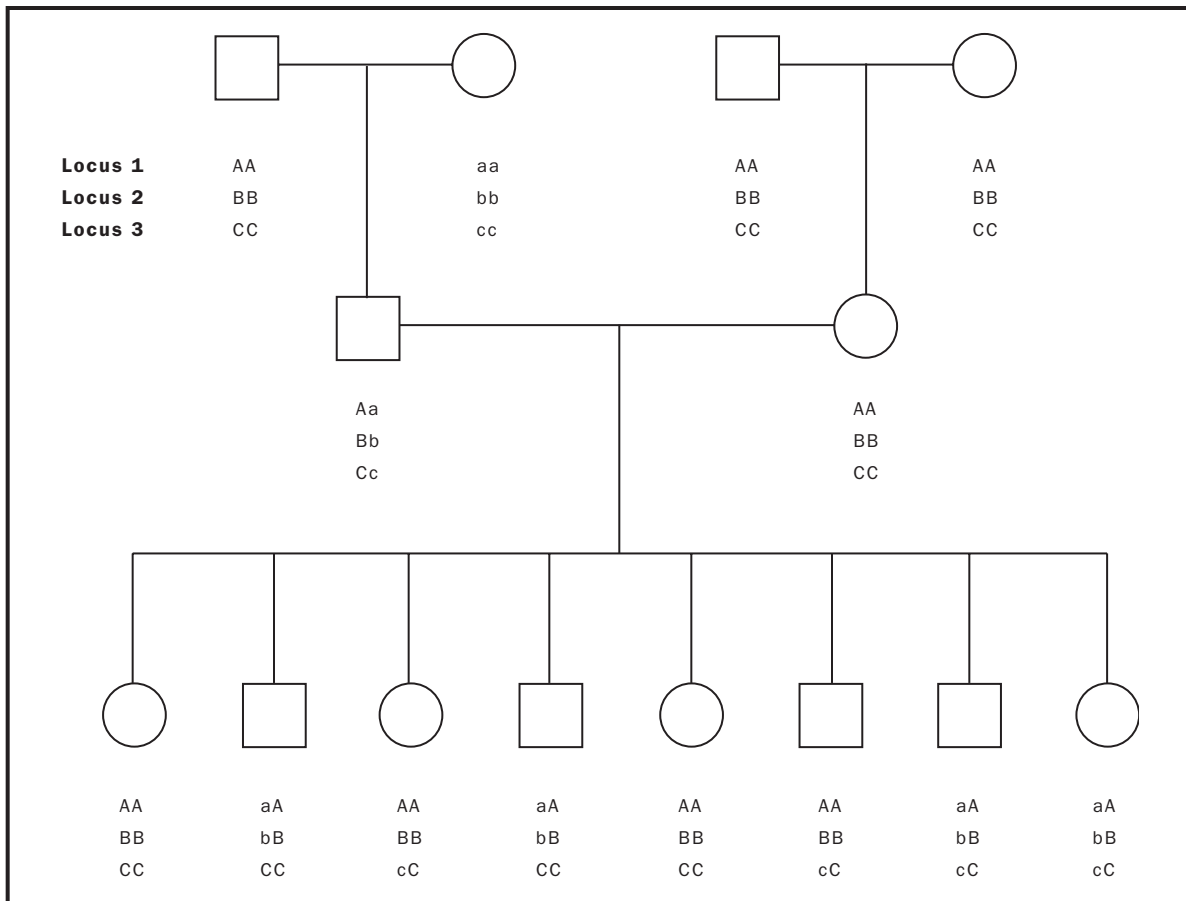
An example of a CEPH family is shown in Figure 1. Using the father as an example (although in other families this could easily occur in the mother), allele *a* at locus 1 and allele *b* at locus 2 are always inherited together. Similarly, allele *A* at locus 1 and allele *B* at locus 2 are inherited together. There has been no recombination between locus 1 and locus 2, and therefore these loci are likely to be close together. In contrast, allele *a* at locus 1 and allele *c* at locus 3 are only inherited together half the time. There have been several recombination events between them, and therefore these loci are likely to be far apart.

The actual distance between two loci is measured using the recombination fraction, which is just the number of recombination events divided by the total number of events that are looked at. In the family diagrammed in the figure, the recombination fraction between locus 1 and locus 2 is 0 recombination events divided by 8 total events, or $0 \div 8 = 0$. The recombination fraction between locus 1 and locus 3 is 4 recombination events divided by 8 total events, or $4 \div 8 = 0.50$. Recombination fractions can vary between 0.00 and 0.50. To generate a complete genetic map of a chromosome, a large number of markers (between 50 and 200, depending on the size of the chromosome) are genotyped in many families, and more complex statistical analyses are used to compare the inheritance across all markers.

There is an additional complication in the analysis of recombination events. The further apart two loci are, the more likely it is that two recombination events could occur between them. The first event will shuffle the alleles, but the second event will reshuffle the alleles back to the way they were. Thus it will look like there were no recombination events when in fact there were two.

Another complication arises from the fact that the occurrence of one recombination event on a chromosome tends to inhibit the occurrence of a second recombination event, especially in regions close to the first one. This is called “interference” and will generally make the map smaller. To account for this, “map functions” have been created that are used to better estimate the true recombination distance between two markers.

Map functions are mathematical equations that are based on assumptions about how much recombination and how much interference exists on a chromosome. Map function distances are measured in units called centimorgans, named for Thomas Hunt Morgan, the first person to develop the techniques of genetic mapping. There are several map functions that have been proposed. Each is named for its originator. The most commonly used



map function is the Haldane map function (named after John Burdon Sanderson Haldane), which assumes that there is no interference between loci. A second map function, the Kosambi map function (named after Damodar Kosambi), assumes a moderate level of interference and seems to more accurately reflect experimental data. Thus the recombination fraction is modified by the map function. Generally the recombination fraction and the centimorgans are very similar for distances from 0.00 to 0.10.

Types of Markers, and Their Advantages and Disadvantages

There are four major kinds of genetic markers that have been used for genetic mapping. The oldest of these is the restriction fragment length polymorphism (RFLP) that was first proposed for genetic mapping in 1980. RFLPs arise from changes in a single **base pair** that can be detected by restriction endonuclease **enzymes**. These enzymes can cut the DNA at that locus if the right base pair is present. Many maps were made with these markers, but they are expensive and time-consuming to genotype, and they generally have only two alleles. Having only two alleles means that in many cases it is impossible to tell the two chromosomes in any person apart for that marker and makes that marker useless for genetic mapping in that family. In the figure, the mother of the eight children has the same alleles at locus 1, the same alleles at locus 2, and the same alleles at locus 3. Thus we cannot tell if there have been any recombination events coming from the

Figure 1. A typical CEPH family with genotypes from three genetic markers. Loci one and two are completely linked to each other, while loci one and three are not linked to each other.

base pair two nucleotides (either DNA or RNA) linked by weak bonds

enzymes proteins that control a reaction in a cell



homologous carrying
similar genes

mother. RFLPs were the first type of marker known to occur almost everywhere, across all the chromosomes.

Variable number of tandem repeat (VNTR) markers were the next markers to be described. These result from the duplication of DNA sequences consisting of 50 to 5,000 base pairs each. The differences between the two **homologous** chromosomes are in the number of repeats present (and thus the length of the locus). These markers are expensive and time consuming to genotype but have the advantage of having many alleles (often more than twenty). Thus almost everyone in the world has a different allele on each paired chromosome at a VNTR locus. This allows more families to give recombination information. Having so many alleles, however, can cause problems, because it can be hard to tell many of the alleles apart during genotyping. VNTRs also tend to occur most often at the ends of chromosomes, not in the middle. This is unlike RFLPs, which occur at all locations on a chromosome.

Microsatellite markers—also known as simple tandem repeat polymorphisms (STRPs), simple sequence repeats (SSRs), or simple sequence length polymorphisms (SSLPs)—have become the most common type of marker for genetic maps. These markers are made of repeats of two, three, or four base pairs, with the variation being the number of repeats. For example, the most commonly used two-base-pair repeat is CA, and the most commonly used four-base-pair repeat is GATA. Thus a microsatellite marker actually varies in length between the paired chromosomes. On one chromosome, there might be eight repeats (CACACACACACACACA), while on the other chromosome there might be ten (CACACACACACACACACACA). Microsatellite markers are easy to genotype and have multiple (three to ten) but usually not large numbers (more than ten) of alleles. They also occur almost everywhere across the chromosome. Most of the genetic maps in use today are made with microsatellite markers.

The most recently described type of marker is the single nucleotide polymorphism (SNP, pronounced “snip”). As the name implies, these are variations at a single base on the chromosome. For example, on some chromosomes a locus might have a C, while on other chromosomes the same locus might have a T. These are the most common markers, with at least three million already described, and seem to occur across the entire genome. As with RFLPs, there are almost always only two alleles at a SNP locus. Individually they suffer the same problem as RFLPs of not being useful in many of the families. They are being used widely now because they are very easy to genotype, are very common (occurring at least ten times more frequently than the other types of markers) and thus can be used in combination with each other.

History of Genetic Mapping

The technique of genetic mapping was first described in 1911 by Thomas Hunt Morgan, who was studying the genetics of fruit flies. Morgan was able to study genetic mapping because he was able to actually see traits in the flies (like having white eyes instead of red) that were caused by mutations in single genes. He noticed that some traits violated Gregor Mendel’s Law of Independent Assortment (which said that any two loci would segregate independently and thus have a recombination fraction of 0.50).

Genetic mapping did not start being applied to humans until the 1950s, because it was hard to know what traits were caused by genetic mutations. When RFLPs were first described in 1980, a large effort was undertaken to generate maps of all the chromosomes. The first such maps were made in the early 1980s but covered only parts of chromosomes and had only a few markers. Maps of whole chromosomes were made by the late 1980s. By the mid-1990s, as the abilities of the research teams improved, and as the statistical methods of analysis were refined, a number of whole-genome (i.e., covering all the chromosomes) genetic maps were generated. These maps were updated and improved, and they were made available on the Internet.

The Comparison of Genetic and Physical Distance

Genetic maps are a measure of distance based on recombination, which is a biological process. A different way of measuring the distance between two loci is to measure the actual number of base pairs between the loci. This is known as the physical distance, and, when many such distances are put together, it makes a physical map.

Genetic maps and physical maps are similar in that the loci will be in the same order. There is also a general correspondence of distance, in that bigger genetic distances usually correspond to bigger physical distances. The overall rule of thumb is that one centimorgan of genetic distance is about one million base pairs of physical distance. However, this comparison can vary dramatically across certain parts of chromosomes. In some areas, one centimorgan might be only 50,000 base pairs (e.g., at the ends of chromosomes, where recombination seems to be increased). In other chromosomal areas (e.g., near the centromere), one centimorgan might be five million base pairs. SEE ALSO CROSSING OVER; GENE DISCOVERY; LINKAGE AND RECOMBINATION; MEIOSIS; MORGAN, THOMAS HUNT; POLYMORPHISMS; REPETITIVE DNA ELEMENTS.

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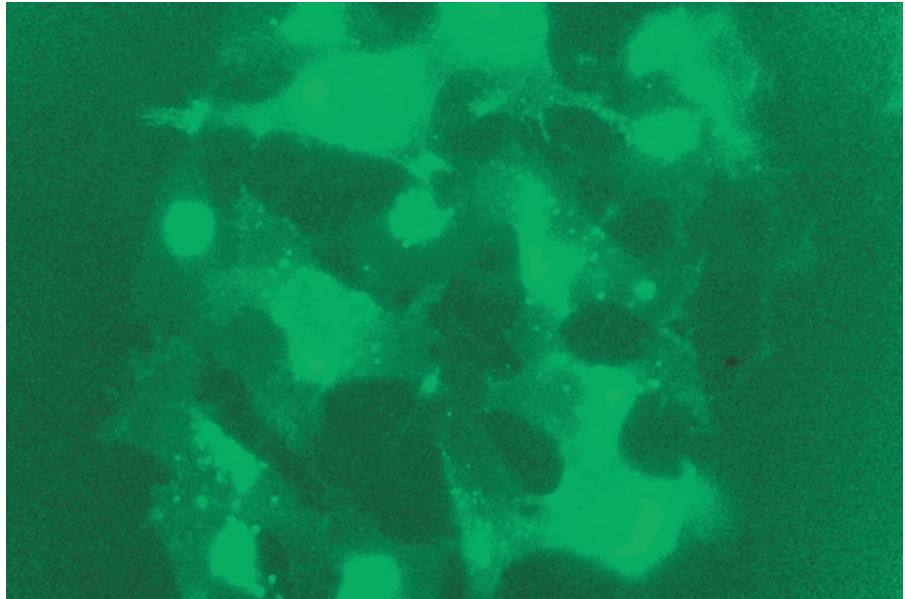
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Marker Systems

Marker systems are tools for studying the transfer of genes into an experimental organism. In gene transfer studies, a foreign gene, called a transgene, is placed into an organism, in a process called transformation. A common problem for researchers is to determine quickly and easily if the target cells of the organism have actually taken up the transgene. A marker allows the researcher to determine whether the transgene has been transferred, where it is located, and when it is expressed (used to make protein).



Green fluorescent protein (GFP) serves as a visible marker in these mouse cells.



phenotype observable characteristics of an organism

The marker itself is also a gene. It is placed next to the transgene to make a single piece of DNA, which is then transferred. Markers are chosen because their gene products (proteins) have obvious effects on the **phenotype** of the organism. If the system is constructed properly, detection of the marker's product indicates that the transgene is present and functioning.

Marker systems exist in two broad categories: selectable markers and screenable markers. Selectable markers are typically genes for antibiotic resistance, which give the transformed organism (usually a single cell) the ability to live in the presence of an antibiotic. Screenable markers, also called reporter genes, typically cause a color change or other visible change in the tissue of the transformed organism. This allows the investigator to quickly screen a large group of cells for the ones that have been transformed. Selectable and screenable markers are essential to genetic engineering in both **prokaryotes** and **eukaryotes**, and are often built into engineered DNA plasmids used for genetic transformation.

prokaryotes single-celled organisms without a nucleus

eukaryotes organisms with cells possessing a nucleus

Selectable Markers

Selectable markers are said to cause either negative or positive selection. Negative selection kills cells that do not have the marker gene, while positive selection kills those that have it but not in the correct place in the chromosome.

Negative selection is most commonly used in the transformation of bacterial cells. A gene for resistance to an antibiotic such as kanamycin is placed on a **plasmid** with the transgene (such as an insulin gene). Resistance genes often code for an enzyme that phosphorylates (adds a phosphate to) the antibiotic, thereby inactivating it. Cells that take up the plasmid can thus tolerate an otherwise lethal exposure to the antibiotic. The researcher exposes the entire group of cells, and harvests those that remain alive.

Positive selection is often performed in mammalian cells grown in cell culture. Because of the complexity of the mammalian cells, it is important

plasmid a small ring of DNA found in many bacteria

that a transgene not only enter the cell, but also be integrated into the correct place in the chromosome. If it does not, it is unlikely to be regulated properly. The “correct place” is the site on the chromosome where the normal gene is found. For example, if the researcher is inserting a human nerve cell gene into a mouse, it should be inserted at the site where the corresponding mouse nerve cell gene sits. Selection of cells with the properly located transgene is accomplished by killing off transformed cells in which the gene is in the wrong place.

This system, an example of positive selection system, has three parts. The first is an antibiotic, the second is an **enzyme** that acts on the antibiotic, and the third is an enzyme that cuts and splices DNA.

The antibiotic ganciclovir is used to kill cells. Ganciclovir is a “nucleotide analog,” meaning it is structurally similar (but not identical) to the building blocks of DNA. It must be phosphorylated before it can be incorporated into DNA in the target cell. Once it is incorporated, it acts like a monkey wrench in the machinery, preventing normal DNA function and thus killing the cell. The enzyme that acts on the ganciclovir is called thymidine kinase (TK). It adds a phosphate on the antibiotic, inactivating the antibiotic. Mammalian TK does not phosphorylate ganciclovir very efficiently, so mammalian cells are not normally killed by it. TK from the *Herpes simplex* virus (HSV) does phosphorylate it efficiently, and any mammalian cell transformed with an active HSV TK enzyme will be killed.

In this system, a plasmid is constructed with the **transgene**, the HSV TK gene, and a “recombination site,” a stretch of DNA that is recognized by the cellular recombinase enzymes that cut and splice DNA. If the transgene is integrated into the chromosome at the site of the normal gene, then the HSV TK gene is eliminated by the cellular “recombinase” enzymes, and the cells are not sensitive to ganciclovir. In improperly transformed cells, the recombinase can’t remove the HSV TK gene, and so those cells will be killed when exposed to ganciclovir.

Screenable Markers

Screenable marker systems employ a gene whose protein product is easily detectable in the cell, either because it produces a visible pigment or because it fluoresces under appropriate conditions. Visible markers rarely affect the studied trait of interest, but they provide a powerful tool for identifying transformed cells before the gene of interest can be identified in the culture. They can also identify the tissues that have (and have not) been transformed in a multicellular organism such as a plant.

Green fluorescent protein (GFP) is used as a screenable marker or a reporter gene in a variety of cells. GFP is a small protein that is isolated from jellyfish. It possesses a trio of amino acids that absorb blue light and fluoresce yellow-green light that is detectable using a fluorescence microscope or other means. Using GFP as a reporter has the enormous advantage that transgenic cells can be located noninvasively, simply by illuminating with blue light and observing the fluorescence. It is a simple protein, and it works in many different model systems (plants, mammalian cell culture, and the like) because it requires no post-translational processing of the protein to make it active. This is helpful, because processing enzymes are typically specific to each type of organism, thus limiting the usefulness of transgenes

enzyme a protein that controls a reaction in a cell

transgenes genes introduced into an organism



in vitro “in glass”; in lab apparatus, rather than within a living organism

hypotheses testable statements

bioinformatics use of information technology to analyze biological data

gel electrophoresis technique for separation of molecules based on size and charge

that require such modifications. In addition, whereas some reporter products are toxic to the cell, GFP is not, and the intensity of the fluoresced light can be used to quantify gene expression.

The *Escherichia coli* bacterium provides another reporter gene system commonly used in plants. The bacterium makes an enzyme, called B-glucuronidase gus A (uid A), that cleaves a group of sugars called B-glucouronides. This enzyme will also cleave a chemical that is added to the culture such that the cleaved chemical is converted into an insoluble, visible blue precipitate at the site of enzyme activity. Many plants lack their own B-glucuronidase enzymes, so it is easy to determine if the plant has been transformed. Enzyme activity can be easily, sensitively and cheaply assayed **in vitro**, and can also be examined in tissues to identify transformed cells and tissues. The level of gene expression can be measured by the intensity of the blue color produced. SEE ALSO CLONING GENES; MODEL ORGANISMS; PLASMID; POST-TRANSLATIONAL CONTROL; RECOMBINANT DNA.

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Mass Spectrometry

Mass spectrometry is a technique for separating and identifying molecules based on mass. It has become an important tool for proteomics, the analysis of the whole range of proteins expressed in a cell. Mass spectrometry is used to identify proteins and to determine their amino acid sequence. It can also be used to determine if a protein has been modified by the addition of phosphate groups or sugars, for example. The technique also allows other molecules, including DNA, RNA, and sugars, to be identified or sequenced.

The use of mass spectrometry has greatly aided proteomics. Whereas DNA sequencing is simple and straightforward, protein sequencing is not. The ability to quickly and accurately identify proteins being expressed in a cell allows a range of **hypotheses** to be tested that cannot be approached by simply looking at DNA. For instance, it is possible with mass spectrometry to determine what proteins are expressed in cancer cells that are not expressed in healthy cells, possibly leading to further understanding of the disease and to development of drugs that target these proteins.

Data derived from mass spectrometry is usually analyzed by computer programs that search databases to help identify the analyzed protein. Such tools are the province of **bioinformatics**. The databases are usually located at a centralized institution and are searched via the Internet.

Ionize, Accelerate, Detect

Proteins to be analyzed, such as those from a cell, are first separated and purified. One technique for this is two-dimensional **gel electrophoresis**.



At the Manchester Metropolitan University in 2001, a technician prepares samples to be analyzed by a mass spectrometry machine. Combined with a database of controlled spectra, this machine can aid in the detection of anthrax spores.



Individual proteins form spots on the gel, which can then be cut out individually. Chromatography can also be used. In this technique, a mixture of proteins is separated by being passed through a column containing inert beads, which slow the proteins to different extents based on their chemical properties. Unlike the two-dimensional gel method, chromatography allows continuous (versus batch) processing of cellular samples, which reduces the requirement for handling of samples and speeds up analysis.

Mass spectrometry begins by ionizing the molecules in the target sample—removing one or more electrons to give them a positive charge. Molecules must be charged so they can be accelerated. The principle is the same as that used in a television or fluorescent light bulb: Charged particles are accelerated by being pulled toward something of the opposite charge. In the mass spectrometer, the speed the molecules attain during acceleration is proportional to their mass (actually, their mass-charge ratio). By determining the speed of the molecules, researchers can calculate their mass.

Proteins are ionized in one of two common ways. The first is matrix-assisted laser desorption ionization, or MALDI. The “matrix” that is used is a crystalline structure of small organic molecules in which the protein is suspended. When excited by a laser, the protein is vaporized (“desorbed”) and ionized to a +1 charge. The second method is electrospray ionization (ESI). In this process, the protein is dissolved in a solution, which is sprayed to form a fine mist (it is ionized at the same time). Evaporation of the surrounding solvent eventually leaves the protein by itself. A benefit of the solution method of ESI is that a mixture of proteins can first be separated by chromatography or capillary gel electrophoresis, and then passed on to the ionizer without additional handling, avoiding the labor-intensive two-dimensional gel method.

Following ionization, the protein is accelerated. The most common way to determine mass is with a “time-of-flight” (TOF) tube. Just as its name implies, this tube is used to determine the time of flight of the protein, allowing a simple determination of velocity (velocity = distance / time). The accelerator imparts a known amount of kinetic energy to the molecule. Since kinetic energy = $1/2(\text{mass})(\text{velocity})^2$, the determination of mass is straightforward.

Applications

Identifying Unknown Proteins. Since several different proteins may have the same mass, simply obtaining the mass of the whole protein is not enough to identify it. However, if it is broken into a characteristic set of fragments (called **peptides**), and the mass of each of these is determined, it is usually possible to identify the protein based on its “peptide fingerprint.”

Sequencing Peptides. Peptides can be sequenced by generating multiple sets of fragments and analyzing the differences in masses among them. Removing a single amino acid from a peptide, for instance, will decrease its mass by a specific amount and at the same time create a new, detectable particle with the same mass. Individual amino acids can be identified by their characteristic molecular masses. Mass spectrometry has made protein sequencing much easier than it had been. The traditional method required about twelve hours to sequence a ten-amino acid peptide. Mass spectrometry can do the same job in about one second. The entire protein need not be sequenced to be identified. Often four to five amino acids are enough.

Identifying Chemical Modifications. Chemical modifications to proteins after they are synthesized (called post-translational modifications) are important for regulation. For instance, the addition of a phosphate group (PO_4) is used to turn on or turn off many enzymes. The presence of such groups can be detected by the additional weight they bring. Sugar groups can be detected in the same fashion. SEE ALSO BIOINFORMATICS; HPLC: HIGH-PERFORMANCE LIQUID CHROMATOGRAPHY; INTERNET; POST-TRANSLATIONAL CONTROL; PROTEINS; PROTEOMICS.

Richard Robinson

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peptides amino acid chains

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McClintock, Barbara

Geneticist 1902–1992

Barbara McClintock was one of the most important geneticists of the twentieth century and among the most controversial women in the history of science. She made several fundamental contributions to our understanding of chromosome structure, put forward a bold and incorrect theory of gene regulation, and, late in her career, developed a profound understanding of the interactions among genes, organisms, and environments. She was born on June 16, 1902, in Hartford, Connecticut, the third of four children and the youngest daughter. She grew up in Brooklyn, New York, and in 1919 she enrolled in the agricultural college of Cornell University, where she received all her post-secondary education. She took a bachelor's degree in 1923, a master's in 1925, and a Ph.D., under the direction of the **cytologist** Lester Sharp, in 1927.

McClintock gravitated toward the cytology and genetics of maize, or Indian corn, and by 1929 she was a rising star in her field. Not quite single-handedly, she made possible the “golden age of maize genetics,” from 1929 to 1935. During those years, McClintock published a string of superb papers identifying novel cytological phenomena and linking them to genetic events. Working with Harriet Creighton, she confirmed that chromosomes physically exchange pieces during the genetic phenomenon known as “crossing over.” She was supported by a series of prestigious fellowships, from the National Research Council, the Guggenheim Foundation, and others, that took her from Cornell to the California Institute of Technology, and to Berlin and back. In 1935 she took a faculty position at the University of Missouri in Columbia. She was not happy there, however, and resigned in 1939, despite the apparent imminence of a promotion with tenure.

In 1941 she took a summer position at Cold Spring Harbor on New York's Long Island, at the Carnegie Institution of Washington's Department of Genetics. It was an ideal position for her, with no teaching or administrative duties. Within a year the post became permanent, and she remained there until her death. On arrival, she continued work that she had begun while at Missouri, investigating a phenomenon called the breakage-fusion-bridge (BFB) cycle. This is a repeating pattern of chromosome breakage she had discovered among strains of maize plants grown from X-rayed pollen. In 1944, during an experiment designed to use the BFB cycle to create new mutations, she discovered numerous “mutable” genes: genes that turned on and off spontaneously during development. In the cells of some of these new mutants lay her most important discovery, chromosome segments that move from place to place on the chromosome. That same year, the National Academy of Sciences honored a woman for only the third time in its eighty-year history when it elected McClintock a member.

During the rest of the 1940s McClintock developed a novel theory of how genes could control the development and differentiation of organisms.



Barbara McClintock, having just received the prestigious \$15,000 Lasker Award in 1981 for her many contributions to the field of genetics.

cytologist a scientist who studies cells

The key to the theory was a new type of genetic element, not a gene but a gene-controller, that first appeared in her 1944 BFB experiment. These “controlling elements,” she argued, inhibited or modulated the effects of the genes near them. She proposed that through coordinated movement from gene to gene (transposition) controlling elements executed the genetic program of development, much as the hammers on a player piano execute the program encoded on a piano roll.

Transposition in maize was confirmed immediately and repeatedly by other researchers. Few scientists, however, could accept her notion that the movements were coordinated. After about 1954, McClintock did little more with transposition, but she continued to work on genetic control for the rest of her long career. Her systems grew increasingly complex, and her thinking led her to comparisons between embryology and evolution.

During the 1970s transposition was discovered in bacteria, and its biochemistry was explained in terms of DNA sequences and enzymatic action. Soon transposition was found to be nearly universal in the living world and was linked to medical fields such as cancer, virology, and immunology. McClintock experienced a rare scientific renaissance. Her theories of genetic control, never widely accepted and by this time rejected outright, were forgotten, and she was reborn as the discoverer of transposition. She won, unshared, the 1983 Nobel Prize in physiology or medicine “for her discovery of transposable genetic elements.”

Since then, the experiments of other researchers have provided at least qualified support for even some of her wilder ideas, such as her conception of the **genome** as a “sensitive organ of the cell”; and the idea that any organism has the genetic instructions to make any other. Some of these findings had appeared by the time she died, on September 2, 1992, but it has been the various genome projects, human and otherwise, that have lent the strongest support to McClintock’s dynamic, interactive vision of nature. Had she lived to be one hundred, she might well have been considered for a second Nobel, this time for her insights into the workings of the genome. SEE ALSO CHROMOSOMAL ABERRATIONS; CHROMOSOMAL THEORY OF INHERITANCE, HISTORY; GENE; MAIZE; MUTATION; TRANSPOSABLE GENETIC ELEMENTS.

Nathaniel Comfort

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McKusick, Victor

Geneticist

1921–

Victor Almon McKusick was born in the little town of Parkman in central Maine, on October 21, 1921. He went to Tufts College in Boston from

genome the total genetic material in a cell or organism

1940 to 1943, and then received his medical degree from Johns Hopkins University in 1946. He has remained at Johns Hopkins ever since, rising from a medical student to physician-in-chief of the entire Johns Hopkins hospital and on the way creating the field of medical genetics and changing forever the way medicine is practiced. Due to his interest, energy, determination, and discoveries, McKusick is often called the Father of medical genetics.

McKusick was first interested in heart diseases, but he discovered that many of the patients he saw had family members who also had heart problems, piquing his interest in the possibility that genes might be causing these diseases. His interest in genetic disorders continued, and in 1957 he established the first Department of Medical Genetics in the country. He started studying many different diseases, including those related to heart defects, blood problems, and dwarfism. He helped found the American College of Medical Genetics and the American Board of Medical Genetics.

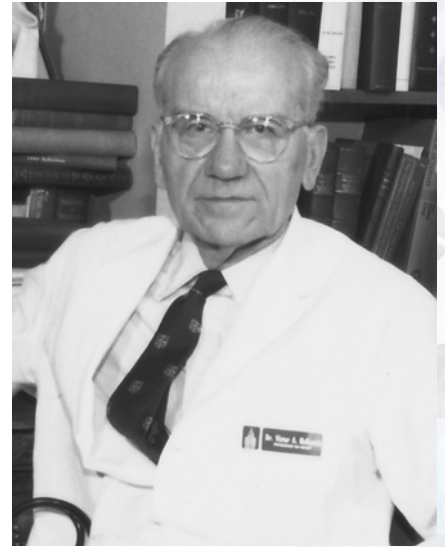
Soon after, McKusick realized that many physicians and scientists knew little about genetics and helped to start one of the first courses in genetics which quickly became known as the “short course in experimental mammalian genetics.” This was a two-week course taught by McKusick and his colleagues from Johns Hopkins and the Jackson Laboratories, a research institute studying the genetics of mice. The course started in 1960 and was taught in Bar Harbor, near the Jackson Laboratories, in McKusick’s home state of Maine. This course quickly became the most well-known and highly respected course in genetics, and brought in students, researchers, and doctors from all over the country. It remains the premier course of its kind today.

When McKusick started the field of medical genetics, there was no list of genetic diseases, nor any one place to go to find information about any of these disorders. He started keeping track of the diseases, and this led to the first edition of his book *Mendelian Inheritance in Man*, which was published in 1966. Even then, when computers were room-sized boxes kept behind closed doors, McKusick knew their value, and he maintained his list using one of these computers. In that first edition, McKusick listed 1,466 different genetic diseases. In the most current edition, over 12,000 diseases are listed. *Mendelian Inheritance in Man* is available on the Internet.

McKusick was one of the first to realize the potential of the Human Genome Project. In 1973 he helped to organize the first Human Gene Mapping Workshop, the predecessor of the Human Genome Project. He helped found the Human Genome Organization (HUGO) in 1988. He has won over twenty prestigious awards, including the Albert Lasker Medical Research Award in 1997, and he has received over twenty honorary degrees from colleges and universities around the world.

McKusick’s accomplishments go well beyond all the papers he has written, all the books he has authored, and all the awards he has won. He has inspired two generations of medical geneticists with his dedication, rigorous scientific approach, warmth, and good humor. SEE ALSO GENETICIST; HUMAN DISEASE GENES, IDENTIFICATION OF; HUMAN GENOME PROJECT.

Jonathan L. Haines



Victor McKusick

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Meiosis

Meiosis is a type of cell division that, in humans, occurs only in male testes and female ovary tissue, and, together with fertilization, it is the process that is characteristic of sexual reproduction. Meiosis serves two important purposes: it keeps the number of chromosomes from doubling each generation, and it provides genetic diversity in offspring. In this it differs from mitosis, which is the process of cell division that occurs in all **somatic** cells.

Overview

All of our somatic cells except the egg and sperm cells contain twenty-three pairs of chromosomes, for a total of forty-six individual chromosomes. This number, twenty-three, is known as the diploid number. If our egg and sperm cells were just like our somatic cells and contained twenty-three pairs of chromosomes, their fusion during fertilization would create a cell with forty-six chromosome pairs, or ninety-two chromosomes total. To prevent that from happening and to ensure a stable number of chromosomes throughout the generations, a special type of cell division is needed to halve the number of chromosomes in egg and sperm cells. This special process is meiosis.

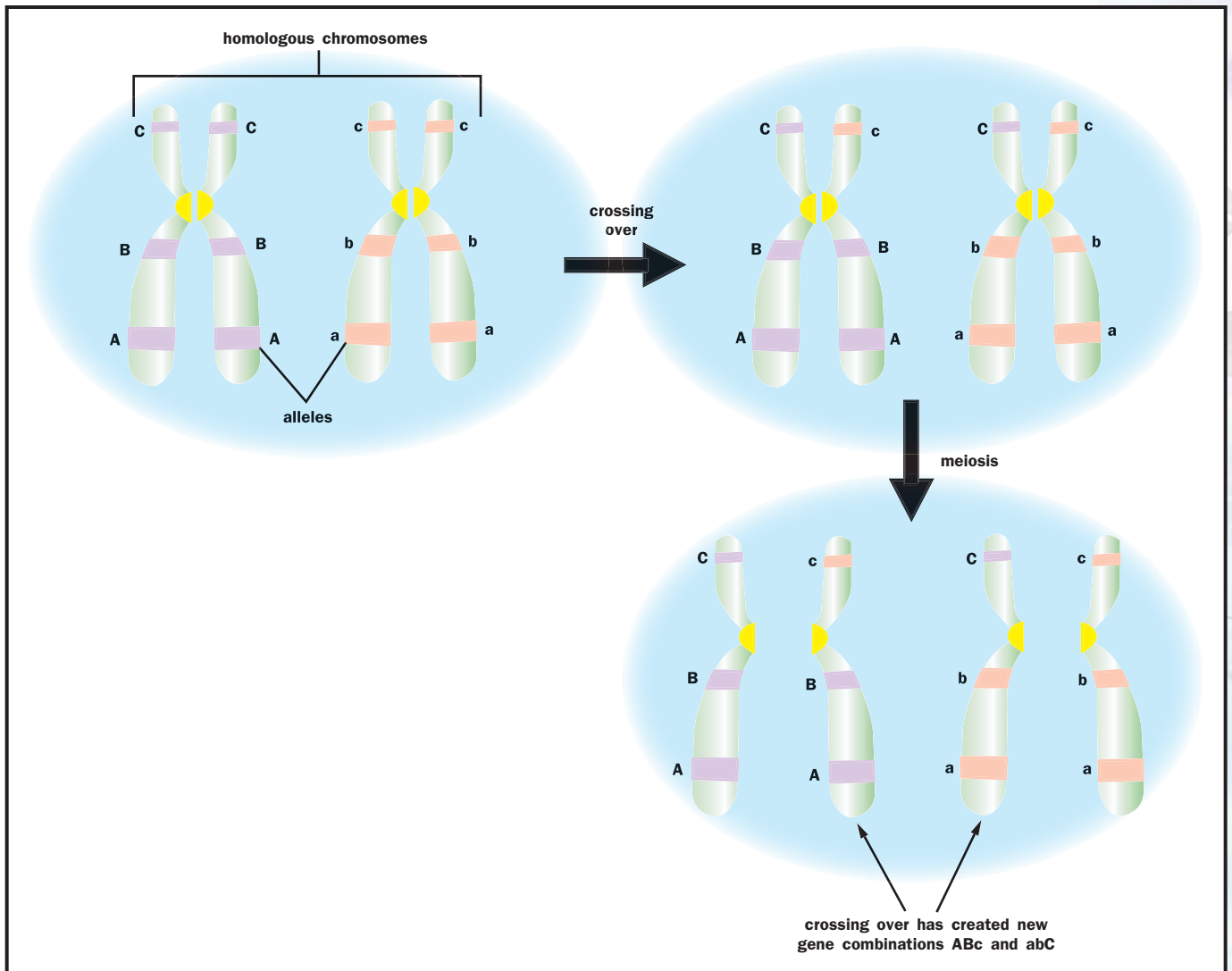
Meiosis creates haploid cells, in which there are twenty-three individual chromosomes, without any pairing. When **gametes** fuse at conception to produce a zygote, which will turn into a fetus and eventually into an adult human being, the chromosomes containing the mother's and father's genetic material combine to form a single diploid cell. The specialized diploid cells that will eventually undergo meiosis to produce the gametes are called primary **oocytes** in the female ovary and primary spermatocytes in the male testis. They are set aside from somatic cells early in the course of fetal development.

Even though meiosis is a continuous process in reality, it is convenient to describe it as occurring in two separate rounds of nuclear division. In the first round (meiosis I), the two versions of each chromosome, called homologues or homologous chromosomes, pair up along their entire lengths and thus enable genetic material to be exchanged between them. This exchange process is called crossing over and contributes greatly to the amount of genetic variation that we see between parents and their children. Subsequently, the two homologues are pulled toward opposite ends of their surrounding cell, thus creating a haploid cluster of chromosomes at each pole, at which point division occurs, separating the two clusters. Meiosis I is therefore the actual reduction division. At the end of meiosis I, each chromosome is still composed of two sister strands (chromatids) held together by a

somatic nonreproductive; not an egg or sperm

gametes reproductive cells, such as sperm or egg

oocyte egg cell



particular DNA sequence of about 220 nucleotides, called the centromere. The centromere has a disk-shaped protein molecule (kinetochore) attached to it that is important for the separation of the sister chromatids in the second round of meiosis (meiosis II). Meiosis II is essentially the same division process as mitosis. Through the separation of the two sister chromatids, a total of four daughter cells, each with a haploid set of chromosomes, are created.

New combinations of alleles result from the crossing over between homologous chromosomes. Adapted from Robinson, 2001.

Meiosis I

Meiosis must be preceded by the S phase of the cell cycle. This is when DNA replication (the copying of the genetic material) occurs. Thus, each chromosome enters meiosis consisting of two sister chromatids joined at the centromere. The first stage of meiosis is a stage called prophase I. First, the DNA of individual chromosomes coils more and more tightly, a process called DNA condensation. The sister chromatids then attach to specific sites on the nuclear envelope that are designed to bring the members of each homologous pair of chromosomes close together. The sister chromatids line up in a fashion that is precise enough to pair up each gene on the DNA

microtubules protein strands within the cell, parts of the cytoskeleton

molecule with its corresponding “sister gene” on the homologous chromosome. This four-stranded structure of maternal and paternal homologues is also called a bivalent.

Next in prophase I is the process of crossing over, in which fragments of DNA are exchanged between the homologous sister chromatids that form the paired DNA strands. Crossing over involves the physical breakage of the DNA double helix in one paternal and one maternal chromatid and joining of the respective ends. Under the light microscope, the points of this exchange can often be seen as an X-shaped structure called a chiasma.

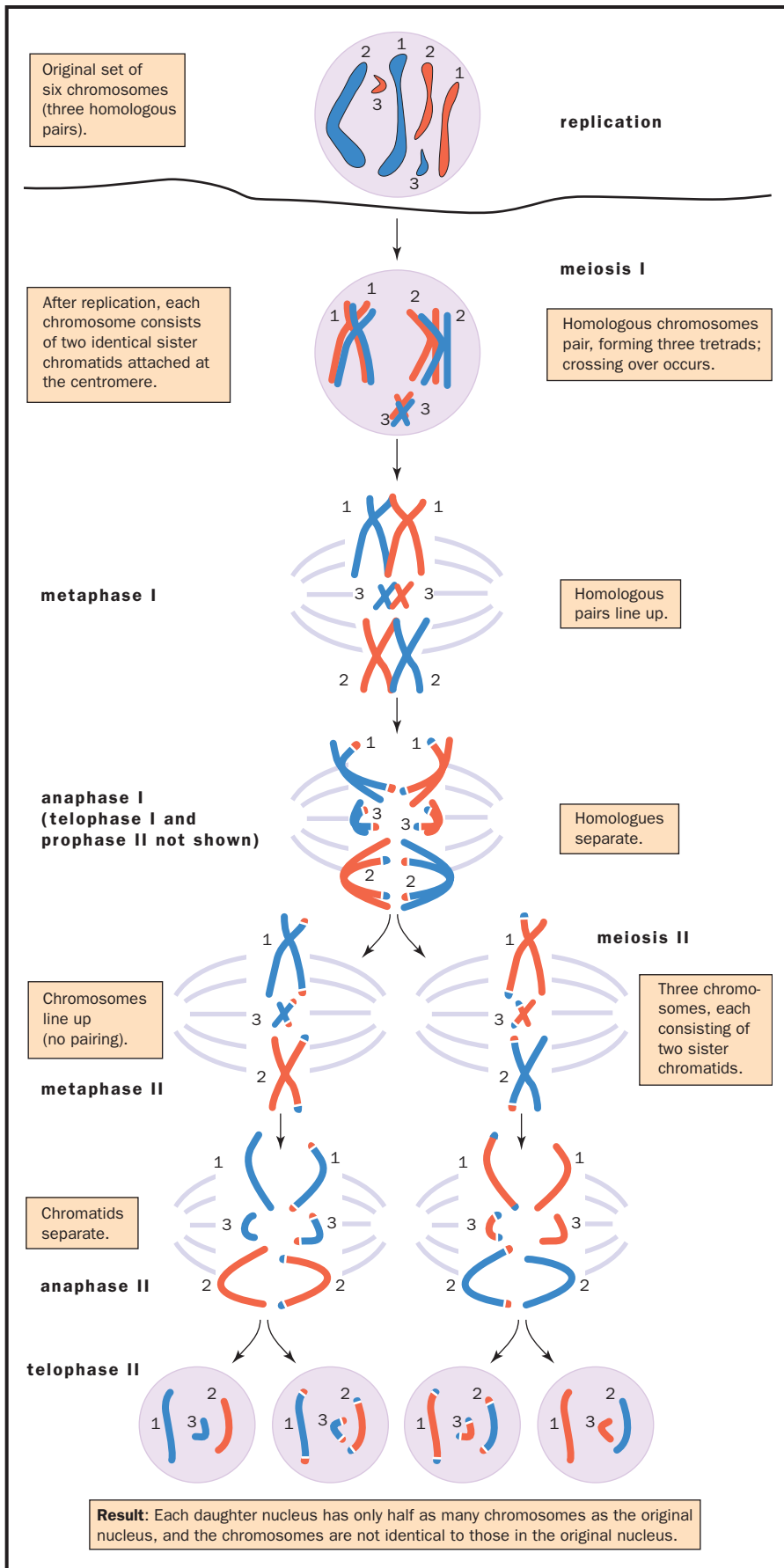
The exchange of genetic material means that new combinations of genes are created on two of the four chromatids: Stretches of DNA with maternal gene copies are mixed with stretches of DNA with paternal copies. This creation of new gene combinations is called “recombination” and is very important for evolution, since it increases the amount of genetic material that evolution can act upon. A statistical technique known as linkage analysis uses the frequency of recombination to infer the location of genes, such as those that increase a person’s risk for certain diseases.

At the beginning of metaphase I, the nuclear envelope has dissolved, and specialized protein fibers called **microtubules** have formed a spindle apparatus, as also occurs in the metaphase of mitosis. These microtubules then attach to the kinetochore protein disks on the two centromeres of the homologous pair of chromosomes. However, there is an important difference between mitosis and meiosis in the way this attachment occurs. In mitosis, microtubules attach to both faces of the kinetochore and thus separate sister chromatids when they pull apart. In meiosis, because the chiasma structures still hold the homologous sister chromatids together, only one face of each kinetochore is accessible to the microtubules. Since the microtubules can only attach to one face of the kinetochore, the sister chromatids will be drawn to opposite poles as a pair, without separation of the individual chromatids.

At the end of metaphase I, the pairs of homologues line up on the metaphase plate in the center of the cell, the spindle apparatus is fully developed, and the microtubules of the spindle fibers are attached to one side of each of the two kinetochores. In anaphase I, the microtubules begin to shorten, thus breaking apart the chiasmata and pulling the centromeres with their respective sister chromatids toward the two cell poles. The centromeres do not divide, as they do in mitosis. At the final stage of meiosis I, called telophase I, each cell pole has a cluster of chromosomes that corresponds to a complete haploid set, one member of each homologous chromosome pair.

The Sources of Genetic Diversity

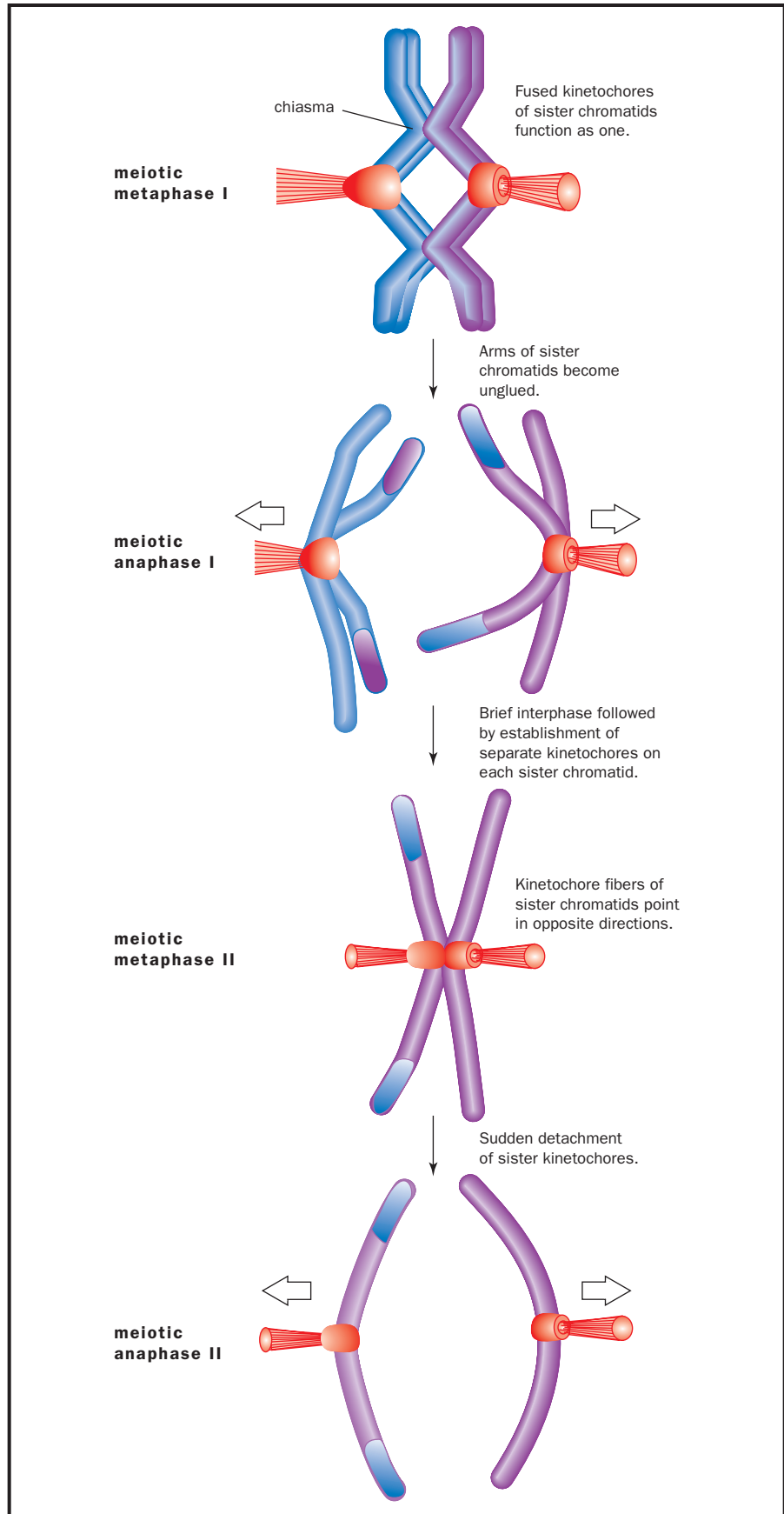
It is completely random whether the maternal or paternal chromosome of each pair ends up at a particular pole. The orientation of each pair of homologous chromosomes on the metaphase plate is random, and a mixture of maternal and paternal chromosomes will be drawn toward the same cell pole by chance. This phenomenon is often called “independent assortment,” and it creates new combinations of genes that are located on different chromosomes. Thus, we have two levels of gene reshuffling occurring in meiosis I. The first occurs during recombination in prophase I, which creates new combinations of genes on the same chromosome. In contrast to mitosis, the sister chromatids of a chromosome are not genetically identical because of the



Meiosis in an organism with six chromosomes. Replication precedes meiosis. Adapted from Curtis, 1994.



Kinetochores fibers separate homologues in Meiosis I, and sister chromatids in Meiosis II. Adapted from Alberts, 1994.



crossing-over process. Anaphase I then adds the independent assortment of chromosomes to create new combinations of genes on different chromosomes. A total of 2^{23} (8.4 million) possible combinations of parental chromosomes can be produced by one person, and recombination further increases this to an almost unlimited number of genetically different gametes.

Meiosis II

Once both cell poles have a haploid set of chromosomes clustered around them, these chromosomes divide mitotically (without reshuffling or reducing the number of chromosomes during division) during the second part of meiosis. This time, the spindle fibers bind to both faces of the kinetochore, the centromeres divide, and the sister chromatids move to opposite cell poles. At the end of meiosis II, therefore, the cell has produced four haploid groups of chromosomes. Nuclear envelopes form around each of these four sets of chromosomes, and the cytoplasm is physically divided among the four daughter cells in a process known as **cytokinesis**.

In males, the four resulting haploid sperm cells all go on to function as gametes (spermatozoa). They are produced continuously from puberty onwards. In females, all primary oocytes enter meiosis I during fetal development but then arrest at the prophase I stage until puberty. During infancy and early childhood, the primary oocytes acquire various functional characteristics of the mature egg cell. After puberty, one oocyte a month completes meiosis, but only one mature egg is produced, rather than the four mature sperm cells in males. The other daughter cells, called polar bodies, contain little **cytoplasm** and do not function as gametes.

Comparison with Mitosis

In summary, the main differences between meiosis and mitosis are that meiosis occurs only in specialized cells rather than in every tissue; it produces haploid gametes rather than diploid somatic cells; and each daughter cell is genetically different from the others due to recombination and independent assortment of homologues, rather than genetically identical. The pairing of homologous chromosomes and crossing over occur only in meiosis.

Chromosomal Aberrations

Meiosis is a very intricate process that requires, among other things, the precise alignment of homologous chromosome pairs and correct attachment of microtubules. During meiosis, errors in chromosome distribution may occur and lead to chromosomal aberrations in the offspring. One example is Down syndrome, where affected children carry three copies of chromosome 21 (trisomy 21). This may be explained by the failure of paired chromosomes or sister chromatids to separate in either sperm or egg, leading to the presence of two copies of chromosome 21. After fertilization with a normal gamete, the zygote will carry three copies, which leads to several **phenotypic** abnormalities, including mental retardation. SEE ALSO CELL, EUKARYOTIC; CHROMOSOMAL ABERRATIONS; CROSSING OVER; DOWN SYNDROME; FERTILIZATION; LINKAGE AND RECOMBINATION; MITOSIS; REPLICATION.

cytokinesis division of the cell's cytoplasm

cytoplasm the material in a cell, excluding the nucleus

phenotypic related to the observable characteristics of an organism

Silke Schmidt